

Learning rate experiments

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can run a controlled comparison of learning rates.

I can explain slow progress oscillation and increasing loss in a toy model.

I can report the limits of a training experiment.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Run a controlled comparison of learning rates

Explain slow progress oscillation and increasing loss in a toy model

Report the limits of a training experiment

Retrieve what you already know

Why can equal epochs give different update counts when batch size changes?

My starting explanation

This week's end goal

By the end, I can explain learning rate experiments using a traced example and a stated limitation.

2. Notice the evidence

Use the same restricted network $h = \text{ReLU}(x)$, $y = vh$, with only v trained. Keep the fictional training pairs (1, 2) and (2, 4), start $v = 0$, and use one full-data batch per update. Compare rates 0.05, 0.2, 0.3, 0.4 and 0.5 for exactly four updates each. Restart at zero for every run.

The training objective is ordinary MSE: $L(v) = 2.5 \text{ times } (v - 2) \text{ squared}$. Its gradient is 5 times $(v - 2)$. A step is new $v = v - \text{rate times } 5 \text{ times } (v - 2)$. This small quadratic lets us explain each result rather than treating the rate as a magic setting.

Rate · v after first update · First training MSE · v after four updates

Rate: 0.05

v after first update: 0.5

First training MSE: 5.625

v after four updates: 1.3671875

Rate: 0.2

v after first update: 2

First training MSE: 0

v after four updates: 2

Rate: 0.3

v after first update: 3

First training MSE: 2.5

v after four updates: 1.875

Rate: 0.4

v after first update: 4

First training MSE: 10

v after four updates: 0

Rate: 0.5

v after first update: 5

First training MSE: 22.5

v after four updates: -8.125

The same rate need not behave this way for another model, input scale, loss convention or batch rule. Rate 0.2 is an exact one-step solution here because of this particular gradient formula.

List the given facts and one fact that is not supplied.

3. See how it works

Write $d = v - 2$, the signed distance from this objective minimum. Substituting the update gives new $d = (1 - 5 \text{ times rate}) \text{ times old } d$. At rate 0.05 the factor is 0.75: distance shrinks without changing sign. At rate 0.3 it is -0.5: distance changes sign and halves. Both can reduce loss.

```
def run(rate, updates=4):  
    v = 0.0  
    history = [v]  
    for _ in range(updates):  
        gradient = 5 * (v - 2)  
        v = v - rate * gradient  
        history.append(v)  
    losses = [2.5 * (value - 2) ** 2 for value in history]  
    return history, losses  
  
for rate in [0.05, 0.2, 0.3, 0.4, 0.5]:  
    values, losses = run(rate)  
    print(rate, values, losses)
```

The helper assumes a finite rate and nonnegative integer update count. Its five logged states include the initial value. At 0.3 the parameter sequence is [0, 3, 1.5, 2.25, 1.875] and loss is [10, 2.5, 0.625, 0.15625, 0.0390625]. Crossing the minimum does not itself mean training failed.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

At rate 0.3 the first update gives $v = 3$, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved divergence?

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Add a rate 1 run from $a = 0$.

Prediction: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across 4 without reducing the initial loss.

5. Words that unlock the idea

Word · Meaning

Word: Learning rate

Meaning: The multiplier controlling the size of a gradient update

Word: Controlled comparison

Meaning: An experiment changing a named factor while holding other stated conditions fixed

Word: Oscillation

Meaning: Repeated movement from one side of a value to the other

Word: Divergence

Meaning: Movement away from a target solution with growing error in this experiment

Use two terms to explain a step in this week's example.

6. Follow a worked example

The update factor explains three behaviours

Rate and factor · Signed distance behaviour · Training loss behaviour

Rate and factor: 0.05 gives 0.75

Signed distance behaviour: Same side and shrinking

Training loss behaviour: Multiplied by 0.5625

Rate and factor: 0.3 gives -0.5

Signed distance behaviour: Alternating and shrinking

Training loss behaviour: Multiplied by 0.25

Rate and factor: 0.5 gives -1.5

Signed distance behaviour: Alternating and growing

Training loss behaviour: Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning rate experiments

Compare learning rates with identical starting loss and update counts.

The facts we will use

Loss $J(a) = (a-4)^2$; gradient $2(a-4)$; start $a = 0$. Compare two updates at rates 0.25 and 0.75.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Learning rate experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Rate0.25

First $a = 2$, loss 4; second $a = 3$, loss 1.



2. Rate0.75

First $a = 6$, loss 4; second $a = 3$, loss 1.



3. Inspect paths

Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

CHANGE ONE THING

Add a rate 1 run from $a = 0$.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Rate0.25

What happens and why: First $a = 2$, loss 4; second $a = 3$, loss 1.

Step: 2. Rate0.75

What happens and why: First $a = 6$, loss 4; second $a = 3$, loss 1.

Step: 3. Inspect paths

What happens and why: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back. Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

What this example does not establish

These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

7. Find and repair a mistake

For a nonzero starting distance, loss strictly decreases on each step of this quadratic when the absolute value of $1 - 5 \text{ times rate}$ is below 1. For positive rates this means $0 < \text{rate} < 0.4$. At 0 the model does not move. At 0.4 it oscillates with unchanged loss. Above 0.4 its distance grows. These bounds belong to this model and objective, not to neural networks generally.

If we switch from mean squared error to the sum over the same two examples, the gradient doubles from $5(v - 2)$ to $10(v - 2)$. Keeping the old rate changes the update. To reproduce the original steps, halve the rate. Likewise, rescaling features can change gradient magnitudes. Record the loss and units alongside the rate.

Locate the earliest unsupported step

Replace MSE by the sum of squared errors on the two training pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

State one repair and the checks you would rerun.

8. Practise together

For rate 0.05, start $v = 0$ and compute two updates and their training MSE values using the lesson objective.

For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?

Explain your prediction to a partner before comparing results.

Practise with less help: Explain a crossing

Use workbook W2 for the connected example “Learning rate experiments”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Learning rate experiments. Proposed explanation: “Crossing the minimum means divergence”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

At rate 0.3 the first update gives $v = 3$, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved divergence?

```
def run(rate, updates=4):  
    v = 0.0  
    history = [v]  
    for _ in range(updates):  
        gradient = 5 * (v - 2)  
        v = v - rate * gradient  
        history.append(v)  
    losses = [2.5 * (value - 2) ** 2 for value in history]  
    return history, losses  
  
for rate in [0.05, 0.2, 0.3, 0.4, 0.5]:  
    values, losses = run(rate)  
    print(rate, values, losses)
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For rate 0.05, start $v = 0$ and compute two updates and their training MSE values using the lesson objective.

Replace MSE by the sum of squared errors on the two training pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

Run A starts $v = 0$ with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final training losses. Can the difference be attributed only to rate?

Propose a repair.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled learning rate experiments evidence sheet to Image-pattern Network Investigation. Show how you can run a controlled comparison of learning rates. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

Record the initial state, data, frozen values, objective, batch rule, update budget, candidate rates, parameter traces and loss traces. These runs compare training behaviour on two examples. They do not establish which setting performs best on independent data. Keep the final test set out of rate selection; later lessons use validation curves to make development choices.

A team says rate 0.2 is the best rate for every neural network because it reached zero training loss here. Replace this claim with a precise experimental conclusion and one limitation.

Next week compares training and validation curves, including when their trends disagree.

Run A starts $v = 0$ with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final training losses. Can the difference be attributed only to rate? Propose a repair.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

The update factor explains three behaviours

Rate and factor · Signed distance behaviour · Training loss behaviour

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Training loss behaviour: Multiplied by 0.5625

Rate and factor: 0.3 gives -0.5

Signed distance behaviour: Alternating and shrinking

Training loss behaviour: Multiplied by 0.25

Rate and factor: 0.5 gives -1.5

Signed distance behaviour: Alternating and growing

Training loss behaviour: Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Compare a new objective independently

Use $J(a) = (a - 1)^2$ with gradient $2(a - 1)$, start $a = 0$ and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use $J(a) = (a - 1)^2$ with gradient $2(a - 1)$, start $a = 0$ and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.